

Solvency Field Theory V12.1

The One-Shatter Closure

A locked shatter-to-recombination toy chain from unified ticking to the local Hubble ledger

Working draft prepared for Shaun Fosmark | 2026-05-22

Supersedes the earlier V12 One Addendum draft and serves as the V12 initial-condition companion to the V11.13 gravity-root document.

Line-1 closure result

With $\alpha = 1/6$ locked, the frozen shatter-to-recombination chain gives $R_{\text{eff,rec}} = 0.628104$ Mly, matching the Gate 11R recombination Hubble-radius ledger $R_{\text{H,rec}} = 0.624698$ Mly to 0.545%.

Claim boundary: toy closure only; not a full CMB/BBN/perturbation derivation and not a derivation of the root invariant or G .

Abstract

This document rewrites the earlier V12 One Addendum around the completed Gate 12A-12L closure sequence. The central proposal is that the missing transition between the pre-spatial unified process and the post-shatter two-scale universe is a lag-burst settlement event: the unified process ticks before ordinary space exists, unresolved phase/settlement demand accumulates, the resulting lag stress shatters the unified state into independent tickers, and the pent-up writes settle as the birth of boundary/spatial meaning. The post-shatter thermal scale a_T controls photon temperature and local reaction clocks, while the settlement-geometry scale $a_L = a_T Q$ controls effective spatial ledger geometry.

Gates 12E-12J established that ordinary thermal cooling from QCD/fm spatialization underreaches local recombination horizon ledgers unless a nonthermal settlement stretch $Q(T)$ is introduced. Gate 12K found a no-cheat source hint: the fitted exponent $\alpha = 0.165105$ satisfies $3\alpha = 0.495315$, within 0.937% of $1/2$. This motivates the locked rule $Q^3 = \sqrt{a_T}$, equivalently $Q = a_T^{1/6}$. Gate 12L then locked $\alpha = 1/6$ and ran the frozen end-to-end chain: boundary-equivalent shatter seed, QCD/fm spatialization, ordinary thermal cooling, and square-root settlement-volume relaxation. The primary frozen candidate gives $R_{\text{eff,rec}} = 0.628104$ Mly, compared with Gate 11R local recombination ledgers $R_{\text{sound}} = 0.431808$ Mly, $R_H = 0.624698$ Mly, and $R_{\text{particle}} = 0.837518$ Mly. The Hubble-radius residual is 0.545%, and the max local target factor error is 1.4546. This is a striking toy closure, not a complete cosmological derivation.

Claim Boundary

This paper is intentionally strict about what is earned and what remains open.

- Earned: a locked end-to-end toy chain from V12 one-process/shatter language to the recombination local Hubble-radius ledger.
- Earned: the need for a two-scale map, a_T for local thermal history and $a_L = a_T Q$ for settlement geometry.
- Earned: a non-fitted candidate exponent $\alpha = 1/6$ from the square-root settlement-volume rule $Q^3 = \sqrt{a_T}$.
- Earned: factor-1.5-level closure against Gate 11R local sound/Hubble/particle ledgers, including a 0.545% match to the recombination Hubble radius.
- Open: a rigorous derivation of $Q^3 = \sqrt{a_T}$ from V11.1B lag/stress diffusion or no-overwrite boundary service dynamics.
- Open: derivation of $N \sim 10^{80}$, the shatter threshold, and the exact QCD/fm spatialization spacing from first principles.
- Open: BBN compatibility, full CMB acoustic spectrum, perturbations, flatness, inflation replacement, and the root invariant / G normalization problem from V11.13.

The document therefore should not be read as a replacement for full cosmology. It is a numerically closed initial-condition toy chain with clearly identified mechanism debts.

1. Version Logic: Why This Is V12.1, Not V11.14

V11.13 is the gravity document. It establishes the weak-field settlement geometry skeleton and reduces the absolute gravitational normalization to one unresolved root invariant. This V12.1 document does not derive that root invariant. Instead, it develops the V12 One initial-condition line: how a unified, pre-spatial process can transition into a post-shatter, spatial, thermally cooling universe with a distinct settlement-geometry scale.

For that reason the recommended title is V12.1 - The One-Shatter Closure. It supersedes the early V12 One Addendum, while remaining a companion to V11.13 rather than a replacement for it.

2. The Three-Sentence Shatter Model

1. The unified process ticks but has no ordinary spatial boundary to write to, so unresolved phase / settlement lag accumulates internally.
2. When accumulated lag stress exceeds the unified state's binding or solvency capacity, the process shatters into independent tickers.
3. The pent-up writes settle as the birth of boundary/spatial meaning, and $Q(T)$ is the continued relaxation of that initial lag burst.

These sentences are the conceptual bridge missing from the earlier V12 One Addendum. Gates 12A-12D describe the seed and spatialization, Gates 12E-12J establish the need for nonthermal settlement stretching and two-scale bookkeeping, Gate 12K identifies the locked rational exponent, and Gate 12L tests the full chain.

3. Core Variables and Scale Separation

The post-shatter model distinguishes the local thermal scale from the effective settlement-geometry scale:

$$a_T(T) = T_{\text{post}} / T$$

$$a_L(T) = a_T(T) Q(T)$$

The thermal scale a_T controls photon temperature, local reaction clocks, and ordinary thermal density scaling. The ledger/settlement scale a_L controls effective spatial settlement geometry. The forbidden identification is $T = T_{\text{post}} / a_L$, because that overcools photons by Q and reintroduces a Q^4 radiation restoration penalty.

Gate 12J casts the allowed bookkeeping as settlement-density relaxation:

$$\rho_{\text{gamma}} \sim a_T^{-4}$$

$$\rho_L \sim a_L^{-3} = (a_T Q)^{-3}$$

Gate 12K motivates the locked source-hint rule:

$$Q^3 = \sqrt{a_T}$$

$$Q = a_T^{1/6}$$

This rule means settlement volume relaxes as the square root of thermal progress. It is a strong hint of diffusive or lag-reservoir behavior, but it is not yet a derivation from V11.1B.

4. Gate Ledger: 12A Through 12L

The following ledger summarizes the completed sequence.

Gate	Question	Verdict	Key number	Boundary
12A	Unified shatter seed audit	Boundary-equivalent seed viable; ordinary spatial ball impossible.	$R_{\text{area}} = 45.59 \text{ km}$ nominal; $t_{\text{shatter}} = 6.626e-23 \text{ s}$; $kT \text{ density} = 20.7 \text{ TeV}$; $E/\text{piece} = 62.4 \text{ MeV}$.	Convention, causality, and mechanism debts remain.
12B	Boundary radius vs spatial radius	Pre-shatter radius must be boundary-equivalent, not ordinary space.	$R_{\text{causal}} = 19.86 \text{ fm}$ while $R_{\text{area}} = 45.59 \text{ km}$; ordinary ball fails.	Spatial dimension itself must be born at shatter.
12C	Post-shatter spatialization map	Spatialization requires metric emergence, not particle transport.	$N = 1e80$ at fm spacing gives AU-scale post-shatter geometry.	Mechanism for spatial assignment open.
12D	Spatialization scale selector	QCD/fm 3D bulk spatialization is viable target.	Best candidates in 2-4 fm range; R_{post} of a few AU.	Exact spacing not derived.
12E	Expansion-to-recombination bridge	Pure thermal cooling underreaches local recombination ledgers.	Best QCD branch needs $q \sim 16\text{-}31$ extra radius factor depending on target.	Requires nonthermal metric/settlement channel.
12F	Modest stretch channel audit	A few-times-ten local stretch is numerically viable.	Best single $q = 25.98$; $\ln q = 3.257$; $\alpha \sim 0.165$.	Mechanism not derived.
12G	Smooth nonthermal stretch law	Smooth $Q(T)$ laws work; constant α works best.	$Q(1 \text{ MeV})=2.12$; $Q(1 \text{ eV})=20.75$; $Q_{\text{rec}}=25.98$.	Law shape viable, not mechanism.
12H	Expansion-rate compatibility	Q must not be naive local Friedmann expansion.	If misread: $H/H_T = 1.165$ and ΔN_{eff} proxy = 2.66.	BBN/CMB rate risk if misinterpreted.
12I	Ledger-vs-thermal scale separation	Two-scale contract formally passes.	a_T controls thermal clocks; $a_L = a_T Q$ controls settlement geometry.	Mechanism for Q open.
12J	Settlement conservation law audit	Q can be booked as settlement-density relaxation.	At rec: $Q^3 = 1.754e4$; $Q^3 = 5.70e-5$; forbidden $Q^4 = 4.558e5$.	Bookkeeping only.
12K	Lag-burst origin of Q	$\alpha = 1/6$ is a strong no-cheat source hint.	$3 \alpha_{\text{fit}} = 0.495315$ within 0.937% of $1/2$; $Q = a_T^{(1/6)}$ predicts q within 3.13%.	Derive square-root rule from V11.1B/no-overwrite.
12L	End-to-end locked closure	Locked $\alpha=1/6$ chain closes at factor-1.5 toy level.	$R_{\text{eff,rec}} = 0.628104 \text{ Mly}$; Hubble radius residual 0.545%.	Full cosmological derivation still open.

5. The Lag-Burst Shatter Picture

Before shatter, there is no ordinary spatial arena into which a tick can be written. The safe wording is not that energy is created or absorbed, but that an attempted write remains as unresolved phase or settlement liability. The process continues ticking, and the unresolved settlement demand accumulates as lag. In SFT language, this is the extreme boundary condition of solvency: demand exists while ordinary spatial capacity is not yet available.

The shatter occurs when the accumulated lag stress exceeds the unified process's ability to remain unified. The moment of shatter is therefore not an explosion of particles through pre-existing space. It is a transition from one unresolved process to many independent write sites, causing pent-up write demand to acquire boundary/spatial meaning.

In this reading, Q(T) is not an add-on expansion. It is the relaxation tail of the initial spatial assignment. Thermal cooling proceeds on a_T; settlement geometry relaxes on a_L = a_T Q.

6. Numerical Spine of the Locked Chain

The primary locked Gate 12L chain uses the frozen Gate 12F/12G spatialization candidate, then locks alpha = 1/6. No alpha fitting or resweeping is performed.

Stage	Symbol	Value	Role
Pre-shatter marker	N	1.0e80	Baryon/tick count marker, not derived here
Boundary-equivalent seed radius	R_area	45.5937 km	Pre-geometric, not ordinary spatial ball
Shatter cycle time	t_shatter	6.626e-23 s	V12 One/Gate 12A marker
Primary spacing	s	2.080201 fm	Frozen QCD/fm spatialization candidate
Post-shatter radius	R_post	4.003905 AU	From frozen primary spatialization
Post-shatter thermal marker	kT_post	94.8596 MeV	Frozen primary candidate
Recombination thermal marker	kT_rec	0.256219 eV	Gate 11R/12L marker
Thermal growth	a_T	3.702279e8	T_post/T_rec
Locked settlement stretch	Q	26.7965	a_T^(1/6)
Settlement volume ratio	Q^3	1.92413e4	sqrt(a_T)
Thermal-only radius at recombination	R_thermal,rec	0.0234398 Mly	Underreaches local ledgers
Effective settlement radius	R_eff,rec	0.628104 Mly	R_post a_T Q

7. Gate 12L Target Comparison

Gate 12L compares the locked prediction to the local recombination ledgers established in Gate 11R. The half-percent hit is specifically to the local Hubble-radius ledger, not to the full last-scattering shell or the present observable universe.

Gate 11R local ledger	Target	12L locked prediction	Factor error
Sound horizon	0.4318077 Mly	0.6281044 Mly	1.45459
Hubble radius	0.6246982 Mly	0.6281044 Mly	1.00545
Particle horizon	0.8375182 Mly	0.6281044 Mly	1.33341

The primary result is a striking toy closure:

$$R_{\text{eff,rec}} / R_{\text{H,rec}} = 1.005452$$

$$\text{Hubble-radius residual} = 0.545\%$$

The max factor error over the sound, Hubble, and particle ledgers is 1.4546. This is why the Gate 12L verdict is STRIKING_TOY_CLOSURE_WITHIN_FACTOR_1P5.

8. Frozen-Candidate Robustness

The primary candidate is best, but the QCD/fm family broadly lands in the local recombination window. This is important: the closure does not rely on one candidate alone.

Frozen candidate	Spacing	R_eff,rec	Max factor error	Status
Primary thermal-Compton fixed	2.0802 fm	0.628104 Mly	1.4546	within 1.5
1 fm	1.0000 fm	0.573153 Mly	1.4612	within 1.5
QCD 200 MeV	0.9866 fm	0.572186 Mly	1.4637	within 1.5
Proton radius	0.8409 fm	0.560870 Mly	1.4932	within 1.5
Classical electron radius	2.8179 fm	0.652394 Mly	1.5108	within 2
Piece-energy reduced Compton	3.1615 fm	0.661845 Mly	1.5327	within 2
Thermal piece fixed	3.6349 fm	0.673489 Mly	1.5597	within 2

9. Interpretation: What the Half-Percent Hubble Hit Means

The Hubble-radius match is the strongest numerical result in this branch. It should be stated precisely. The locked chain does not claim to predict the full observable universe, the full CMB angular spectrum, or the last-scattering shell. It predicts an effective recombination settlement radius and compares it to local causal ledgers at recombination. The most direct match is to the Hubble radius: 0.628104 Mly versus 0.624698 Mly.

The result is nevertheless striking because it emerges from a long chain with a locked rational exponent: unified process language, boundary-equivalent seed, QCD/fm spatialization, ordinary thermal cooling, square-root settlement-volume relaxation, and Gate 11R local horizon comparison.

The clean paper sentence is:

With $\alpha = 1/6$ locked by the square-root settlement-volume rule $Q^3 = \sqrt{a_T}$, the frozen V12 shatter-to-recombination toy chain maps a QCD/fm post-shatter spatialization scale to $R_{\text{eff,rec}} = 0.628$ Mly, matching the Gate 11R recombination Hubble-radius ledger to 0.545%, while remaining within factors 1.45 and 1.33 of the sound and particle ledgers.

10. Forbidden and Allowed Claims

Forbidden claims:

- Do not claim the full root invariant or G has been derived.
- Do not claim $\alpha = 1/6$ is rigorously derived until the V11.1B/no-overwrite derivation exists.
- Do not claim BBN, perturbations, flatness, or the full CMB spectrum have been solved.
- Do not call Q ordinary adiabatic expansion, ordinary particle transport, or photon cooling.
- Do not identify $T = T_{\text{post}}/a_L$ or $\rho_{\gamma} \sim a_L^{-4}$.
- Do not call the pre-shatter radius an ordinary spatial radius.
- Do not call the local Hubble-radius match a prediction of the present observable universe size.

Allowed claim:

- A locked V12 shatter-to-recombination toy chain closes at the factor-1.5 level against local recombination horizon ledgers, with a 0.545% residual against the local Hubble radius.
- The chain uses a non-fitted rational exponent $\alpha = 1/6$ suggested by square-root settlement-volume relaxation.
- The result provides a strong mechanism target: derive $Q^3 = \sqrt{a_T}$ from SFT settlement lag dynamics.

11. The Derivation Program After This Closure

Discovery branch status: closed at toy level. Derivation branch status: open. The next work is not another sweep; it is mathematical physics. The target is to derive the square-root settlement-volume rule from existing SFT structures:

- V11.1B lag diffusion / settlement-stress dynamics.
- No-overwrite / no-rewrite boundary service constraints.
- Finite ledger capacity during the transition from one process to many write sites.
- Lag reservoir relaxation after the initial burst of unwritten phase.
- The relation between thermal progress a_T and settlement volume progress Q^3 .

The desired derivation target is:

$$Q^3 = \sqrt{a_T}$$

$$d \ln Q / d \ln a_T = 1/6$$

A successful derivation would turn Gate 12K from a source hint into a mechanism. Until then, $\alpha = 1/6$ is locked as a no-cheat candidate, not a proven theorem.

12. Conclusion

The V12 One Addendum originally proposed that the universe could begin as one unified process whose accumulated writes become spatial only at shatter. Gates 12A-12L now turn that picture into a locked toy chain. The key addition is the lag-burst transition: before ordinary space, write demand cannot be externally settled; lag accumulates; shatter creates independent write sites; $Q(T)$ is the relaxation tail of the initial spatial assignment.

The final locked chain is compact:

one process -> lag-burst shatter -> QCD/fm spatialization -> a_T -> $Q = a_T^{1/6}$ -> $R_{\text{eff}, \text{rec}}$

Numerically, the primary frozen candidate yields $R_{\text{eff}, \text{rec}} = 0.628104 \text{ Mly}$, with a 0.545% residual against the Gate 11R local recombination Hubble radius. This is the strongest initial-condition closure in the V12 line. It does not finish the theory, but it sharply identifies the next derivation target: square-root settlement-volume relaxation from SFT lag dynamics.

Appendix A. Reference Scripts and Captures

- SFT_V12_GATE_12A_unified_shatter_seed_audit.py

- SFT_V12_GATE_12B_boundary_radius_vs_spatial_radius_audit.py
- SFT_V12_GATE_12C_post_shatter_spatialization_map.py
- SFT_V12_GATE_12D_spatialization_scale_selector.py
- SFT_V12_GATE_12E_expansion_to_recombination_bridge.py
- SFT_V12_GATE_12F_modest_metric_stretch_channel_audit.py
- SFT_V12_GATE_12G_nonthermal_metric_stretch_law_MOBILE.py
- SFT_V12_GATE_12H_expansion_rate_compatibility_audit.py
- SFT_V12_GATE_12I_ledger_vs_thermal_scale_separation_audit.py
- SFT_V12_GATE_12J_settlement_conservation_law_audit.py
- SFT_V12_GATE_12K_lag_burst_settlement_origin_audit.py
- SFT_V12_GATE_12L_end_to_end_shatter_to_recombination_closure_alpha_locked.py
- SFT_V12_Line_1_Capture_Gates_12E_to_12J.md

Appendix B. One-Line Status

V12.1 closes the discovery branch at toy level: the frozen shatter-to-recombination chain with $\alpha = 1/6$ locked reaches the recombination local Hubble ledger to 0.545%, while leaving the derivation of the square-root settlement law, the root invariant, and full cosmological constraints open.